HEET SONI

AI Engineer | Software Engineer | Full Stack AI Developer

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PROFESSIONAL SUMMARY

AI & Data Science undergraduate with hands-on experience building AI agents, LLM applications, Retrieval-Augmented Generation (RAG) systems, Computer Vision solutions, and Full-Stack Software. Developed end-to-end AI applications through academic, internship, and personal projects, with a focus on scalable backend systems and practical AI solutions.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C++

AI Technologies: LLMs, AI Agents, Retrieval-Augmented Generation(RAG), Computer Vision, Natural Language Processing(NLP), Prompt Engineering

AI Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, OpenCV, YOLOv8

Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, WebSockets

Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3

Databases: PostgreSQL, MongoDB, MySQL, ChromaDB, SQLite

Developer Tools: Git, GitHub, Docker, Linux

EDUCATION

B.Tech in Artificial Intelligence and Data Science | A. D. Patel Institute of Technology (ADIT), CVM University, Anand, Gujarat, India

- CGPA: 8.05 / 10 (Till 6th Semester) *June 2023 – June 2027(Expected)*

PROJECTS

VAJRA - Foundation Language Model Framework | Python • PyTorch • Transformer Architecture • Distributed Trainings • LLMs *May 2026 - Present*

- Built an open-source framework for training, evaluating, and packaging decoder-only Transformer language models, with custom BPE tokenization and reproducible training pipelines.
- Implemented LLaMA-style components including RoPE, RMSNorm, SwiGLU, and Grouped-Query Attention, alongside dataset sharding, mixed-precision training, checkpointing, and automated evaluation.

MoodLens - Real Time Multimodal Emotion AI | Next.js • React • TypeScript • TensorFlow.js • ONNX Runtime Web • Transformers.js • Tailwind CSS

September 2026 – October 2026

- Developed a browser-based emotion analysis platform supporting facial expression recognition, text analysis, batch photo processing, video analysis, and voice-based emotion comparison.
- Implemented client-side AI inference and an interactive mood dashboard with emotion timelines, session journaling, visual analytics, and JSON/CSV exports, keeping media processing local to the browser.

TrafficIQ - Autonomous Multi-Agent Traffic Intelligence System | Python • FastAPI • YOLOv8 • ByteTrack • ZeroMQ • WebSockets

May 2026 – July 2026

- Developed a fully offline edge-native traffic intelligence platform using computer vision and multi-agent coordination to replace fixed-timer traffic signals with adaptive real-time control.
- Engineered a distributed FastAPI system integrating YOLOv8, ByteTrack, ZeroMQ, WebSockets, and deterministic traffic algorithms for real-time vehicle tracking, emergency routing, and live traffic analytics.

AutoPitch Studio | Python • FastAPI • Whisper • Ollama • Groq API • Celery • Redis • Node.js • PptxGenJS *July 2026 – Aug 2026*

- Developed an AI-powered presentation generation platform that converts spoken ideas into fully editable Microsoft PowerPoint decks using local speech recognition and LLM-driven content generation.
- Engineered a modular pipeline integrating Whisper, Ollama/Groq, Celery, Redis, and PptxGenJS to support privacy-first processing, asynchronous task execution, and native PPTX generation.

WORK EXPERIENCE

Full-Stack Software Developer| XTIN Services LLP | Software Development • Backend Engineering • Full-Stack Development *Oct 2025 – Oct 2026*

• Developed full-stack application features spanning frontend interfaces, backend logic, database operations, authentication, and payment workflows.

• Integrated external services, resolved software bugs, and supported application deployment and maintenance to improve functionality and reliability.

Artificial Intelligence Intern | Codec Technologies | Python • NumPy • Scikit-learn • TensorFlow • NLP • Machine Learning *May 2026 – June 2026*

• Developed machine learning and deep learning applications, including a Multilayer Perceptron (MLP) implemented from scratch using NumPy for handwritten digit recognition and an NLP-based spam email classifier using TF-IDF and Support Vector Machines.

• Built interactive AI applications by visualizing neural network activations, evaluating multiple classification models, and applying feature engineering, model training, and performance optimization on real-world datasets.

AI-Assisted Full Stack Web Development | Graphic Era Deemed to be University | AI-Assisted Development • Full-Stack Development • Debugging • Software Engineering *June 2026 – August 2026*

• Developed AI-assisted full-stack web solutions, applying modern AI tools for code generation, debugging, optimization, and rapid problem solving across frontend and backend development.

• Built and improved web-based applications by combining AI-assisted workflows with conventional programming, translating technical requirements into functional solutions and resolving development challenges.

CERTIFICATIONS

- Google Machine Learning Crash Course — **Google**
- Build Your Own Small Language Model — **Google DeepMind**
- Hewlett Packard Enterprise – Software Engineering Job Simulation — **Forage**
- Datacom – Automation AI Accelerator: From Co-pilot to Autonomous Agent — **Forage**
- Vibe2Ship – India's Biggest Vibe Coding Hackathon — **Coding Ninjas**
- BCG – GenAI Job Simulation — **Forage**

TECHNICAL COMPETITIONS

• **PromptWars Virtual (Hack2Skill)** — Ranked #86 / 43,260 (Top 0.2%) in the AI Code Submission Challenge.

• **Google's AI Agents: Intensive Vibe Coding Capstone** — Built and deployed Agentic AI solutions using modern LLM workflows.

• **DoraHacks Casper Agentic Buildathon** — Developed AI agent applications in a competitive hackathon environment.

• **Skill-to-Agent Dual Cascade Hackathon (DoraHacks)** — Developed reusable AI agent skills and autonomous agent workflows in a competitive global hackathon focused on the AI Agent ecosystem.

• **Vibe2Ship 2026 (Coding Ninjas)** — Participated in India's largest Vibe Coding Hackathon.